Trapezium Rule (Numerical Integration)

Exam code: 9MA0

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Help me understand

Trapezium rule (numerical integration)

What is the trapezium rule?

- The **trapezium rule** is a numerical method of integration used to find the **approximate area** under a curve
- It can be used when **analytical methods** of **integration** won't work
- It finds an **approximation** of the area by finding the **sum of the areas** of trapeziums beneath the curve
- The **formula** for an **estimate** of the area when splitting it into *n* strips (trapeziums) is

$$\int_a^b y\,dx \approx \frac{1}{2}h\{(y_0+y_n)+2(y_1+y_2+\ldots+y_{n-1})\} \text{ where}$$

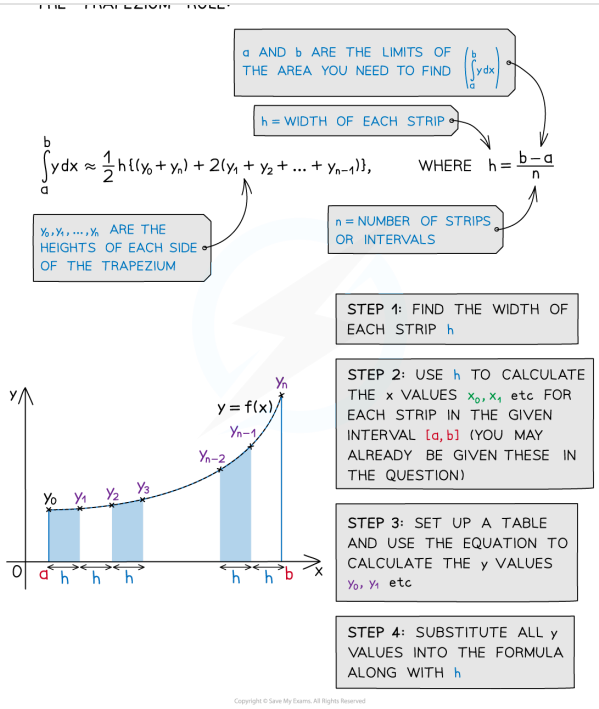
$$h=\frac{b-a}{n}$$

- If you have ***n* strips** then you need ***n* + 1 *y*-values**

Stuck on something?

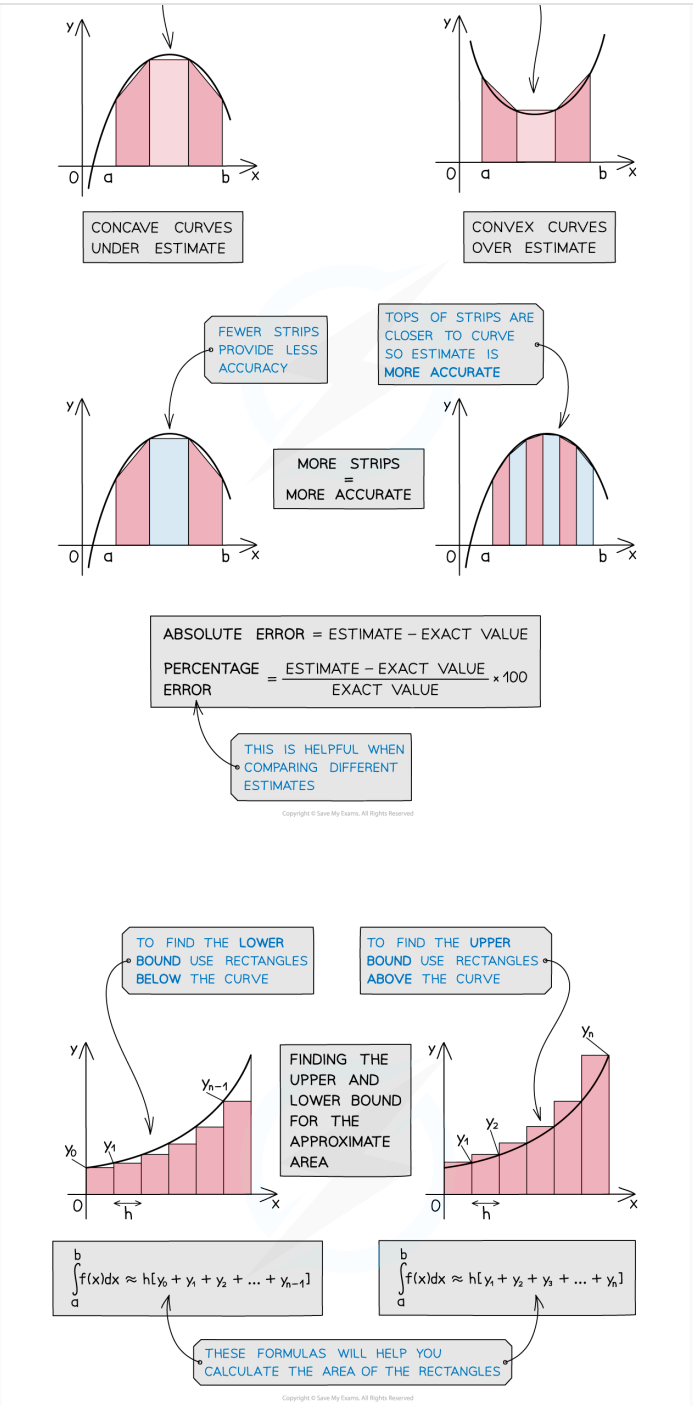
Ask questions and get help with exactly what you're revising.

Got it



How do I know if it is an underestimate or an overestimate?

- Depending on the shape of the curve, the trapezium rule could be an **underestimate** or **overestimate**
- The **more trapezium strips** used the **more accurate** the estimate
- Calculating the **percentage error** can tell you how **accurate** your estimate is
- Using **rectangular strips** can find the **upper and lower bounds** of the interval for the trapezium rule



Examiner Tips and Tricks

The formula for the trapezium rule is given in the formula booklet.

Worked Example

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